

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

Annual Examination 2026 — ANSWER KEY (Version 1)

Biology — Class 9 (Science Group)

Syllabus: Chapter 1 — The Science of Biology | Chapter 2 — Biodiversity | Chapter 3 — The Cell | Chapter 4 — Cell Cycle | Chapter 5 — Tissues, Organs and Organ Systems | Chapter 6 — Molecular Biology | Chapter 7 — Metabolism (Sections 7.1 to 7.6 only)

Total Marks: 36

Time: 1 Hour 45 Minutes

Paper: Exam-16 (Version 1)

SECTION A — Answers to Multiple Choice Questions (20 × 1 = 20 Marks)

Q1. Fraction of 100 healthy persons with NO trace of *Plasmodium* in their blood.

Correct answer: **B) 93/100** 1 mark

The textbook's malaria experiment states that of 100 healthy persons examined, 7 had *Plasmodium* in their blood (in the incubation period), so the remaining 93 out of 100 had no trace of it at all. This is the exact data-analysis example the chapter gives for ratio and proportion.

Source in her textbook: Ch 1, §1.8 Cause of Malaria – Experiments & Results, p.14 — SLO 8 (steps of the biological method; malaria case study data)

Q2. A statement that describes an observable occurrence in nature which appears always true, but does not explain why, is called a:

Correct answer: **D) scientific law** 1 mark

A **scientific law** is a statement that describes an observable occurrence in nature which appears to always be true; it does not explain WHY something happens, only THAT it happens. A **scientific theory** (option C, the wrong twin) goes further and EXPLAINS the underlying mechanism, based on facts that have been repeatedly confirmed through observation and experiment. A hypothesis is only a tentative, testable idea, and deduction is a reasoning step, not a finished statement.

Source in her textbook: Ch 1, §1.7 Hypothesis, Theory and Law, Summary point 13, p.13 — SLO 9 (evaluate hypothesis, theory and law)

Q3. Bioinformatics is a combination of biology and:

Correct answer: **A) information technology** 1 mark

The textbook describes **bioinformatics**, one of biology's emerging careers, as a combination of biology and information technology that helps understand complex biological data.

Source in her textbook: Ch 1, §1.4 Careers in Biology, p.10 — SLO 6 (identify careers in biology, including emerging careers)

Q4. Correctly written scientific name of the lion, *Panthera leo*, following binomial nomenclature rules.

Correct answer: **C) *Panthera leo*** 1 mark

The textbook's rules for binomial nomenclature require the genus name to start with a capital letter, the species name to be entirely in lower case, and the whole name to be italicized when printed (or underlined by hand). Only *Panthera leo* follows all three rules together.

Source in her textbook: Ch 2, §2.7 Binomial Nomenclature, p.29 — SLO 7 (rules for writing binomial names)

Q5. Multicellular heterotrophic eukaryotes lacking a cell wall and chlorophyll, including both insects and fish/frogs:

Correct answer: **A) Animalia** 1 mark

The textbook defines **kingdom Animalia** as multicellular, heterotrophic eukaryotes that lack cell wall and chlorophyll and can generally move from place to place; it includes invertebrates (e.g. insects, starfish) and vertebrates (e.g. fish, frogs, man).

Source in her textbook: Ch 2, §2.4 The Domain Eukarya – Kingdom Animalia, p.25 — SLO 4 (three domains and the kingdoms of Domain Eukarya)

Q6. Which is NOT one of the three components of the cytoskeleton?

Correct answer: **D) Ribosomes** 1 mark

The cytoskeleton is made up of three types of protein fibres: **microtubules**, **microfilaments** and **intermediate filaments**. Ribosomes are a completely separate organelle (the site of protein synthesis) and are not part of the cytoskeleton.

Source in her textbook: Ch 3, Cytoskeleton, p.37 — SLO 3 (sketch and outline the roles of subcellular organelles)

Q7. Layer of the plant cell wall formed after the cell stops growing, thickened with lignin:

Correct answer: **B) Secondary wall** 1 mark

The **secondary wall** is formed inside the primary wall in some cells (e.g. xylem vessels) only after the cell has stopped growing; it is thick and rigid due to the presence of lignin. The **primary wall** is thin and made of cellulose fibres present while the cell is still growing, and the **middle lamella** is the sticky pectin layer that holds neighbouring cell walls together.

Source in her textbook: Ch 3, Cell Wall, p.33 — SLO 3 (sketch and outline the roles of subcellular organelles, including the cell wall)

Q8. Pinching-in structure formed during cytokinesis in an ANIMAL cell:

Correct answer: **C) cleavage furrow** 1 mark

In an animal cell, the cell membrane invaginates (pinches inward) at the equator to form a **cleavage furrow**, which continues to grow inward until the membranes join and separate the two daughter cells. A **phragmoplast** (option B, the wrong twin) is instead the structure formed by spindle fibres at the equator of a PLANT cell during cytokinesis — it is not part of animal cell division.

Source in her textbook: Ch 4, §4.3 Cytokinesis, p.52 — SLO 2 (explain mitosis and its stages, by use of sketch and diagrams)

Q9. A cell with $2n = 14$ undergoes meiosis. Chromosomes per cell at the end of MEIOSIS I:

Correct answer: **A) 7** 1 mark

Meiosis I separates the homologous chromosome pairs, halving the chromosome number. A cell that starts with $2n = 14$ chromosomes therefore produces two cells with $n = 7$ chromosomes each at the end of meiosis I (each chromosome still has 2 sister chromatids at this point, but the CHROMOSOME NUMBER is already halved).

Source in her textbook: Ch 4, §4.4 Meiosis, p.54–56 — SLO 2/3 (explain meiosis and its stages; maintenance of chromosome number)

Q10. A cell with $2n = 10$ replicates its DNA (S phase). Chromosomes/chromatids at late prophase of MITOSIS:

Correct answer: **D) 10 chromosomes, 2 chromatids each** 1 mark

DNA replication during the S phase does not change the number of chromosomes — it doubles the amount of DNA in each chromosome, giving each chromosome 2 identical sister chromatids joined at the centromere. So a $2n = 10$ cell still has 10 chromosomes at late prophase of mitosis, each now made of 2 chromatids.

Source in her textbook: Ch 4, §4.2 S Phase / §4.3 Prophase, p.48, 50 — SLO 1/2 (cell cycle; stages of mitosis)

Q11. Skeletal system's role in homeostasis, besides support/protection/movement:

Correct answer: **C) Calcium** 1 mark

Table 5.2 states the skeletal system (bones, cartilage, joints, tendons, ligaments) also helps maintain homeostasis by regulating the level of **calcium** and other minerals in the blood.

Source in her textbook: Ch 5, Table 5.2 Major Organ Systems, p.69–70 — SLO 6 (discuss how organs/systems of the body work to maintain homeostasis)

Q12. Organ system made of leukocytes, tonsils, adenoids, thymus and spleen, that removes disease-causing microbes:

Correct answer: **A) Immune system** 1 mark

Table 5.2 lists the **immune system's** organs/structures as leukocytes, tonsils, adenoids, thymus and spleen, with its homeostatic role being to remove disease-causing microbes from the body. The **lymphatic system** (lymph, lymph nodes, lymph vessels) is a separate, related system that maintains tissue fluid balance.

Source in her textbook: Ch 5, Table 5.2 Major Organ Systems, p.69–70 — SLO 6 (discuss how organs/systems of the body work to maintain homeostasis)

Q13. Muscular system (with the integumentary system) helps maintain homeostasis by:

Correct answer: **D) regulating body temperature** 1 mark

Table 5.2 states the muscular system (skeletal, cardiac and smooth muscles) works together with the integumentary system (skin, hair, nails) to help regulate body temperature, besides providing movement, support and protection.

Source in her textbook: Ch 5, Table 5.2 Major Organ Systems, p.69–70 — SLO 6 (discuss how organs/systems of the body work to maintain homeostasis)

Q14. Nitrogen base with a DOUBLE-ring structure:

Correct answer: **B) Guanine** 1 mark

Guanine is a purine, and purines (adenine and guanine) have a double-ring structure. **Cytosine, thymine and uracil** (options A, C, D) are all pyrimidines, which have only a single-ring structure.

Source in her textbook: Ch 6, §6.7 Nucleic Acids – Nucleotide structure, p.87 — SLO 7 (structure of DNA as a double helix made of nucleotides)

Q15. General property of ALL lipids (insolubility in water), distinct from "saturated/unsaturated" (which applies only to some fatty acids):

Correct answer: **A) hydrophobic** 1 mark

The textbook states lipids are "all hydrophobic in nature," meaning they are insoluble in water but soluble in organic solvents — this applies to the WHOLE class of lipids. **Saturated** (option C, the wrong twin) instead describes a type of FATTY ACID that has all C–C single bonds; it is a separate, narrower idea that applies only to some lipid components, not to lipids as a whole.

Source in her textbook: Ch 6, §6.4 Lipids, p.82 — SLO 4 (outline the structure, function and sources of lipids)

Q16. Approximate number of different amino acid types commonly found in proteins:

Correct answer: **C) 20 to 25** 1 mark

The textbook states there are about 20–25 different types of amino acids commonly found in proteins, each with a different alkyl (R) group that gives it distinct chemical properties.

Source in her textbook: Ch 6, §6.3.2 Amino Acids, p.79 — SLO 3 (outline the structure of proteins with the structure of amino acids)

Q17. Covalent bonds between the phosphate groups of ATP, which store releasable energy:

Correct answer: **B) High-energy bonds** 1 mark

The textbook states the covalent bonds between the phosphate groups of ATP are called "high energy" or "energy-rich" bonds, represented in diagrams by a wavy line; the energy stored in these bonds can be used by the cell to carry out its functions.

Source in her textbook: Ch 7, §7.6.2 Structure of ATP Molecule, p.101 — SLO 6 (ATP as the energy currency of the cell)

Q18. What happens to an enzyme molecule after it converts substrate into product:

Correct answer: **D) It regains its original shape and can be reused for the same reaction again** 1 mark

The textbook states that although an enzyme's structure may slightly change during a reaction, it regains its actual shape and remains unchanged at the end of the reaction, so it can be used repeatedly — a very small quantity of an enzyme is capable of catalysing a huge amount of substrate.

Source in her textbook: Ch 7, §7.2.1 Characteristics of Enzymes – Enzymes are required in small quantity, p.95 — SLO 2 (enzymes as biological catalysts and their characteristics)

Q19. Series of connected reactions where each enzyme acts on the product of the previous one:

Correct answer: **C) metabolic pathway** 1 mark

The textbook defines a **metabolic pathway** as a series of connected chemical reactions where multiple intermediate molecules are produced before an initial substrate is converted into a final product; many enzymes work together in a specific sequence to build such a pathway.

Source in her textbook: Ch 7, §7.2.1 Characteristics of Enzymes – Enzymes make metabolic pathway, p.95 — SLO 2 (enzymes as biological catalysts and their characteristics)

Q20. Digestion of food into smaller, simpler molecules is an example of:

Correct answer: **B) catabolism** 1 mark

Biochemical reactions that break down complex molecules into simpler molecules are called catabolic reactions or **catabolism**; the textbook gives digestion as an example, where larger, complex food molecules are broken down into smaller, simpler molecules.

Source in her textbook: Ch 7, §7.1 Metabolism and Its Types, p.94 — SLO 1 (metabolism, catabolism and anabolism)

SECTION B — Short Questions (2 × 4 = 8 Marks)

2(i). Trace a polypeptide's folding journey through the four levels of protein structure. 4 marks

Primary structure: a linear polypeptide chain with a specific number, type and sequence of amino acids, joined by peptide bonds. 1

Secondary structure: the polypeptide folds upon itself, due to additional (hydrogen) bonds, into a higher structural level such as a coil or pleated sheet. 1

Tertiary structure: further folding of the secondary structure, with even more extra bonds, giving the protein its overall 3-D (often globular) shape. 1

Quaternary structure: two or more polypeptides, each already folded into a tertiary structure, bonded together to form one functional protein. 1

(As the textbook notes, a protein does not remain in its primary structure after synthesis — it folds into higher levels to perform its specific role; the fibrous or globular shape of a protein depends on its structural level.)

Source in her textbook: Ch 6, §6.3.3 Structure of Proteins, Fig. 6.4, p.79–80 — SLO 3 (outline the structure, function and sources of proteins)

2(ii). Why is it impossible to eradicate malaria? Give at least four reasons. 4 marks

Any four of the following reasons (1 mark each), all grounded in the biology of *Plasmodium* and its mosquito vector as discussed in the chapter:

1. *Plasmodium* has an **incubation period** during which an infected person shows no symptoms at all (as in the 7 out of 100 "healthy" persons in the textbook's own experiment who were carrying the parasite silently) — such hidden carriers keep the disease circulating without anyone realising it. 1

2. Different species of *Plasmodium* cause fevers that repeat on different cycles (every 24, 48 or 72 hours), so no single fixed treatment schedule can catch every case. 1

3. The mosquito vector (*Anopheles*) breeds in stagnant, marshy water, which is extremely difficult to drain or eliminate completely, especially across a whole country. 1

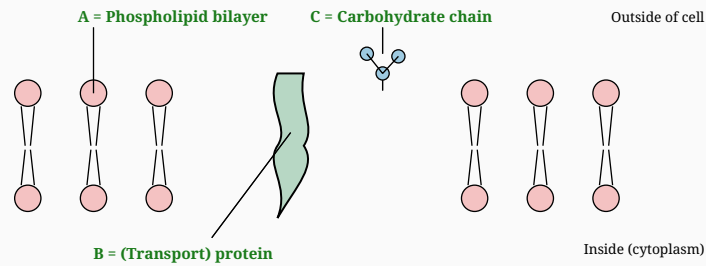
4. As long as both the parasite *Plasmodium* and its mosquito vector exist, and mosquitoes can bite an infected person and then a healthy person, the disease keeps being transmitted in a continuous cycle — breaking the cycle needs vector control AND drug treatment to work together at the same time across a whole population, which is very hard to achieve completely. 1

Source in her textbook: Ch 1, §1.8 Cause and Spread of Malaria, p.13–14, and Exercise Section II Q7, p.19 — SLO 8 (biological method; malaria case study)

SECTION C — Long Question (1 × 8 = 8 Marks)

3. Fluid mosaic model of the cell membrane. 8 marks

(a) Labels 3



A = Phospholipid bilayer. 1 B = Protein (transport protein embedded in the membrane). 1 C = Carbohydrate chain. 1

(b) Chemical composition 2

Chemically, the cell membrane is composed of proteins **60–80%**, phospholipids **20–40%**, and traces of carbohydrates. 2

(c) The fluid mosaic model 3

The fluid mosaic model postulates that the cell membrane consists of a double layer (bilayer) of phospholipids in which proteins are incorporated in a mosaic fashion. 1 The protein molecules float like icebergs in a sea-like fluid of phospholipids, meaning they are not fixed in place but can move within the lipid bilayer. 1 Eukaryotic cell membranes also contain cholesterol, which prevents the membrane from stiffening and is required for the fusion of secretory vesicles with the membrane. 1

Source in her textbook: Ch 3, Cell Membrane / Plasma Membrane, Fig. 3.4, p.33 — SLO 3 (sketch and outline the roles of subcellular organelles, including the cell membrane)

— End of Answer Key —